Parag Sarvoday Sahu

Senior Undergraduate | Electrical Engineering with Minors in Computer Science and Engineering
3D Computer Vision | Inverse Rendering | Neural Rendering | Computational Imaging

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Education

Indian Institute of Technology Gandhinagar — **9.02**/10
B.Tech in Electrical Engineering with Minors in Computer Science and Engineering — 2022-2026

Chhattisgarh Public School, Raipur — Percentage: **95.8**
Class XII, Central Board for Secondary Education — 2020-2021

Chhattisgarh Public School, Raipur — Percentage: **94**
Class X, Central Board for Secondary Education — 2018-2019

Publications

TensoIS: Feed-Forward Tensorial Inverse Subsurface Scattering for Perlin Distributed Heterogeneous Media
Ashish Tiwari, Satyam Bhardwaj, Yash Bachwana, **Parag Sarvoday Sahu**, Shanmuganathan Raman
Pacific Graphics 2025 (CGF Journal Track)
Project Page | DOI: 10.1111/cgf.70242

Research Experience

IndiaAI Fellowship: Inverse Rendering with Gaussian Primitives

Independent Research | IndiaAI | IIT Gandhinagar — *Jan '26 - Present*

- Awarded the competitive IndiaAI Fellowship to pursue independent research on inverse rendering.
- Developing a Multi-View Photometric Stereo (MVPS) framework using Gaussian primitives for accurate reconstruction of geometry and surface normals.
- Designing differentiable rendering and optimization pipelines to disentangle material properties and complex illumination from multi-view image observations.

Research Internship, 3DVisLab

Advisor: Prof. Avinash Sharma • IIT Jodhpur • Blog — *July '25 - Dec '25*

- Developed learning-based frameworks for reflective symmetry detection in 3D shapes, enabling symmetry-aware representations for compression and analysis.
- Implemented mesh processing and deep learning techniques for high-fidelity 3D human head reconstruction.

Real-Time 3D Gaussian Splatting for Novel View Synthesis

Computer Graphics | 2nd Place, 3D Gaussian Splatting Challenge, SIGGRAPH Asia 2025 — *Oct '25 - Nov '25*

- Designed a coarse-to-fine training strategy with progressive resolution scaling for efficient optimization.
- Achieved strong PSNR within a strict 60-second training budget.
- Secured 2nd place globally and received an invitation to present at SIGGRAPH Asia 2025 in Hong Kong.

Inverse Rendering of Heterogeneous Translucent Objects

Computer Vision & Graphics | Prof. Shanmuganathan Raman | IIT Gandhinagar — *Aug '24 - June '25*

- Estimated subsurface scattering parameters of heterogeneous translucent media from multi-view RGB images using a feed-forward inverse rendering pipeline.
- Generated a large-scale synthetic dataset using Mitsuba 3 with heterogeneities modeled via fractal Perlin noise.
- Captured real-world objects and environment maps to evaluate generalization beyond synthetic training data.
- **Resulted in a co-authored publication accepted to the Computer Graphics Forum (Pacific Graphics 2025).**

Summer Research Internship, Photonic Sensors Lab

SRIP, IIT Gandhinagar

Advisor: Prof. Arup Lal Chakraborty • IIT Gandhinagar — *May '24 - Jun '24*

- Designed a mobile methane sensing system using optical spectroscopy and FPGA-based signal processing.
- Implemented a lock-in amplifier on FPGA hardware for precise signal extraction under noise.
- Developed SPI-based communication between FPGA and Raspberry Pi for real-time data acquisition.

In-Band Full Duplex Radios with Self-Interference Cancellation

Adaptive Filtering | Prof. Nithin V. George | Presentation — *Jan '24 - Apr '24*

- Studied full-duplex communication systems and self-interference cancellation techniques through literature review.
- Implemented steepest descent–based adaptive filtering algorithms in MATLAB for interference suppression in batch and online settings.

- Evaluated robustness under noise, observing performance degradation in non-Gaussian environments.

Selected Projects

3D Gaussian Splatting Renderer and Training Pipeline

Computer Graphics | IIT Gandhinagar *Oct '25 - Nov '25*

- Implemented a simplified 3D Gaussian rasterizer in PyTorch, including 3D-to-2D projection, covariance transformation, and differentiable splatting.
- Developed alpha and transmittance computation for ordered Gaussians, and blended colour, depth, and silhouette maps through accumulated transmittance.
- Trained isotropic 3D Gaussian representations from posed multi-view images (toy truck dataset), achieving high PSNR/SSIM on held-out views.

Scene Describer for the Visually Impaired

Embedded Systems & AI Integration | Prof. Jhuma Saha | IIT Gandhinagar | Project Link *Mar '25 - Apr '25*

- Built a low-cost assistive system to capture and audibly describe scenes for visually impaired users using AI.
- Integrated ESP32-CAM, Azure AI Vision, and ESP8266 for image captioning and audio playback.
- Developed a Python controller for image retrieval, AI captioning, speech synthesis, and audio streaming.

Awards and Achievements

- Secured **2nd place** globally in the **3D Gaussian Splatting Challenge**, SIGGRAPH Asia 2025.
- **Dean's List**, 6th semester - awarded to the top 5% of students in a discipline for academic excellence (Official Listing).
- Awarded the **Bipin and Rekha Shah Scholarship** for academic and overall excellence at IIT Gandhinagar (Official Listing).
- Awarded the **Prof. DV Pai Scholarship** for academic and overall excellence at IIT Gandhinagar (Official Listing).
- Ranked in the **top 1%** among over one million candidates in **JEE Advanced 2022** for admission to the IITs.

Skills

Languages: Python, C/C++, MATLAB, Verilog
Frameworks: PyTorch, NumPy, Pandas
Tools: Mitsuba 3, Vivado, Git, Arduino

Relevant Courses

Computer Vision | **Machine Learning** | Data Structures and Algorithms | Discrete Mathematics | Theory of Computing | Matrix Methods for Signal Processing, Data Science and Machine Learning | Digital Signal Processing | Signals, Systems, and Random Processes | Probability, Statistics, and Data Visualization | Numerical Methods | Data-Centric Computing | Calculus of Single Variable and Linear Algebra